Fonds National de la Recherche Luxembourg

MINIMAL MODEL FOR THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS

Tim Herpich
Université du Luxembourg

October 9, 2018

NECD 2018, Berlin

Introduction and Motivation

- Nonequilibrium phase transitions are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- Stochastic thermodynamics provides a systematic treatment of nonequilibrium thermodynamics $\Longrightarrow$ thermodynamics of nonequilibrium phase transitions.
- Practical motivation to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
  - Limitations of energy-converting nanomachines $\Longrightarrow$ Consider ensembles of nanomachines
  - Phase transitions affect performance of large ensembles of machines?
  - Role of the number of nanomachines?
- Synchronization as a promising candidate for triggering synergy effects in the ensemble.

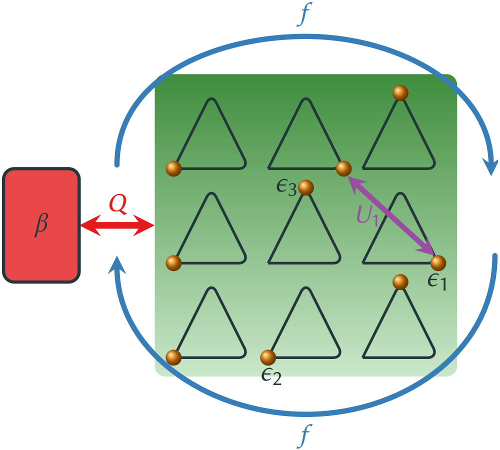
Introduction and Motivation

- Nonequilibrium phase transitions are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- Stochastic thermodynamics provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ thermodynamics of nonequilibrium phase transitions.
- Practical motivation to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
  - Limitations of energy-converting nanomachines $\Rightarrow$ Consider ensembles of nanomachines
  - Phase transitions affect performance of large ensembles of machines?
  - Role of the number of nanomachines?
- Synchronization as a promising candidate for triggering synergy effects in the ensemble.

$\Rightarrow$ Search for a (tractable) thermodynamic model exhibiting synchronization.

Open Three-State Unit Network

- Network of N all-to-all coupled three-state [PRL **96**, 145701 (2006)] units.
- Open system in contact with heat bath.
- Interaction energy $U(\mathbf{N}) = \frac{u}{2N} \sum_{k=1}^{3} N_k^2 + U_0$.
- Subjected to torque f acting as a global bias.



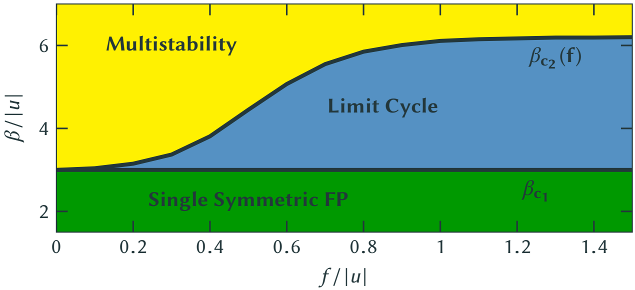
- Markovian master equation with Arrhenius rates satisfying local detailed balance

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'}\, P_{\mathbf{N}'}\,, \qquad \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta\left[\Delta F^{eq}(\mathbf{N},\mathbf{N}') - \sigma(\mathbf{N},\mathbf{N}')f\right], \qquad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} \overbrace{S^{int}(\mathbf{N})}^{\ln \frac{N!}{\prod_i N_i!}}$$

- Mean-field dynamics given by a nonlinear master equation

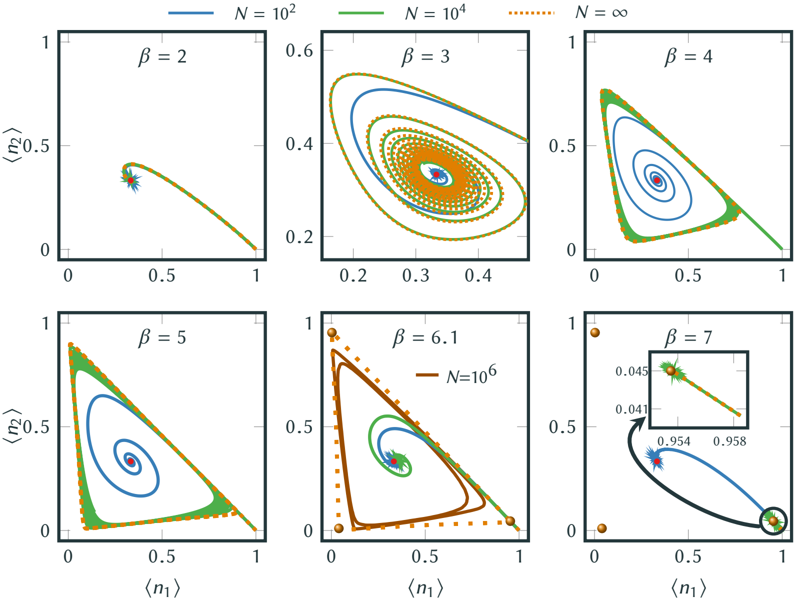
$$\dot{\overline{n}}_i(t) \equiv \langle \overline{n}_i(t) \rangle = \sum_j k_{ij}(\overline{n}_i, \overline{n}_j)\, \overline{n}_j(t), \qquad \ln \frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\overline{n}_i - \overline{n}_j) - \sigma(i,j)f\right].$$

- For $\epsilon_i = \epsilon$, the nonlinear equation has a symmetric fixed point $\dot{\bar{n}}_i = \sum_j k_{ij}\, \bar{n}_j \big|_{\bar{n}_j = 1/3} = 0$.
- Linear stability indicates Hopf bifurcation: $\lambda_\pm = -\Gamma(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\,\Gamma \sinh\left(\frac{\beta f}{2}\right)$.



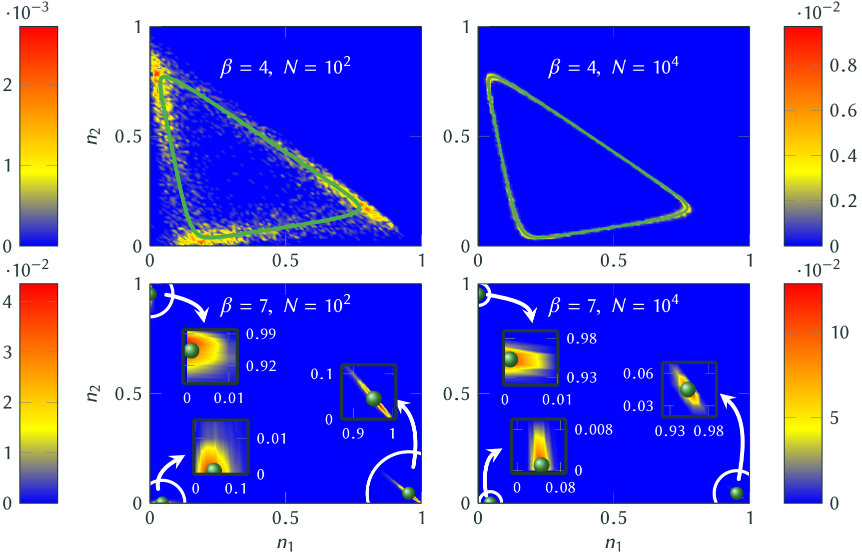
- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c_{1,2}}$.
- Three distinct fixed points in the low-temperature regime ($\beta \gg \beta_c$) Energy-driven
- Symmetric solution in the high-temperature regime ($\beta \ll \beta_c$) Entropy-driven
- Thermodynamic consistency constraints stable oscillations to emerge only at intermediate temperatures ($\beta_{c_1} < \beta \leq \beta_{c_2}$).

$\Longrightarrow$ Minimal thermodynamic model exhibiting synchronization



$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



$\beta = 4$

$N = 10^2$

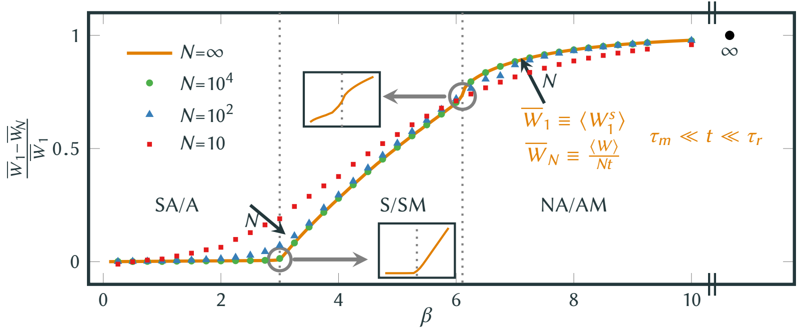
$N = 10^4$

$\beta = 7$

$N = 10^2$

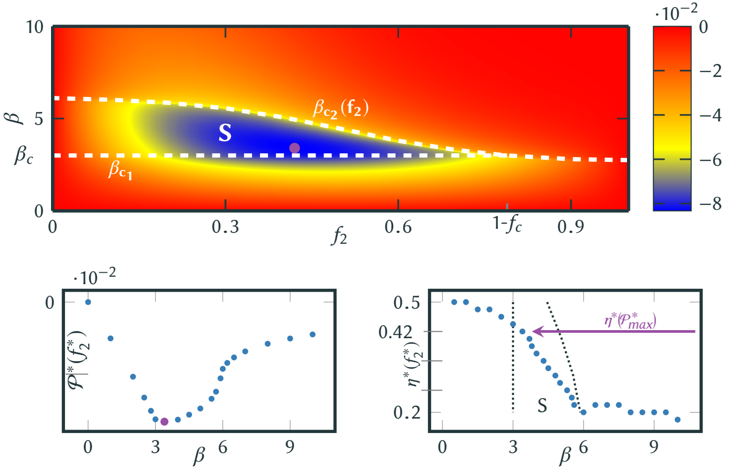
$N = 10^4$

- First Law: $\mathrm{d}_t\langle E\rangle = \langle\dot{Q}\rangle + \langle\dot{W}\rangle$
- With this setup, work needs to be done on average to maintain the system, $\langle\dot{W}\rangle > 0$, which is on average dissipated as heat into the bath, $\langle\dot{Q}\rangle < 0$.



- (Signatures of) Nonequilibrium phase transitions at the critical points (for large but finite N). ["First-order phase transition" at β_{c_1} and "second-order phase transition" at β_{c_2}].
- Reduction of the dissipated work for finite interactions $u < 0$.
- Dissipated work is further reduced for smaller networks in the two high-temperature regimes.

- Work-to-work conversion $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- Maximization of the output power $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2}\mathcal{P}\big|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \frac{f}{f_1}\Big|_{f_2=f_2^*}$



- Power output is maximized in the synchronization phase.
- The efficiency at maximum power, $\eta^*(f_2^*)$, is surprisingly close to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

Summary and Outlook

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

$\Rightarrow$ T.H., Thingna, and Esposito, PRX **8**, 031056 (2018)

Summary and Outlook

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

$\Longrightarrow$ T.H., Thingna, and Esposito, PRX **8**, 031056 (2018)

- Fluctuations? Herpich *et al.*, *In preparation*

Summary and Outlook

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

$\Longrightarrow$ T.H., Thingna, and Esposito, PRX **8**, 031056 (2018)

- Fluctuations? Herpich *et al.*, *In preparation*
- Universal properties in Potts model? Herpich, Esposito, *In preparation*

Summary and Outlook

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

$\Longrightarrow$ T.H., Thingna, and Esposito, PRX **8**, 031056 (2018)

- Fluctuations? Herpich *et al.*, *In preparation*
- Universal properties in Potts model? Herpich, Esposito, *In preparation*
- **Genuine thermodynamic machine in synchrony?** Any ideas would be highly appreciated!

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

$\Longrightarrow$ T.H., Thingna, and Esposito, PRX **8**, 031056 (2018)

- Fluctuations? Herpich *et al.*, *In preparation*
- Universal properties in Potts model? Herpich, Esposito, *In preparation*
- **Genuine thermodynamic machine in synchrony?** Any ideas would be highly appreciated!

Thank you for your attention!